SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT – 1

CSA47 – DEEP LEARNING

CO1 – MATHEMATICAL FOUNDATIONS AND MACHINE LEARNING PRINCIPLES

DOCUMENT REPORT

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Course Code	CSA4708 – Deep Learning

INTRODUCTION

This report analyses the mathematical foundations and machine learning principles required to build a healthcare risk-prediction system. The hospital dataset contains patient attributes such as age, blood pressure, glucose level and cholesterol, together with disease labels for some records and unlabelled records for others.

The report covers feature representation and normalization, vector operations, probability, Bayes' theorem, entropy, distance measurement, bias–variance trade-off, model capacity, supervised learning and unsupervised clustering. The final task proposes a scalable strategy for analysing one million patient records.

TASK 1 – PATIENT DATA REPRESENTATION AND NORMALIZATION

a) Effect of using raw features

The features have different units and numerical ranges. Age may be measured in tens, while blood pressure, glucose and cholesterol may be measured in hundreds. If raw values are used directly, features with larger numerical magnitudes can dominate distance calculations and may produce larger gradients during optimization. This can make learning slower or less stable and can give an unintended importance to a feature simply because of its scale.

b) Feature matrix and min–max normalization

Consider the following illustrative patient data. The values are examples used to demonstrate the calculation.

Patient	Age	BP	Glucose	Cholesterol
P1	25	120	90	180
P2	45	140	110	220
P3	60	160	130	240

The feature matrix is:

X = [[25, 120, 90, 180], [45, 140, 110, 220], [60, 160, 130, 240]].

Apply min–max normalization separately to each column:

$x' = (x - x_{min}) / (x_{max} - x_{min})$

For example, normalized age for P2 = (45 − 25) / (60 − 25) = 20/35 = 0.5714. The normalized matrix (rounded to four decimal places) is:

Patient	Age	BP	Glucose	Cholesterol
P1	0.0000	0.0000	0.0000	0.0000
P2	0.5714	0.5000	0.5000	0.6667
P3	1.0000	1.0000	1.0000	1.0000

c) Importance

Normalization puts features on a comparable scale. Distance-based methods such as K-nearest neighbours and K-means then avoid allowing a large-scale feature to dominate the distance. Gradient-based methods such as logistic regression and neural networks can optimize more reliably because feature scales are more balanced. In practice, normalization parameters should be calculated from the training data and then applied to validation and test data to avoid data leakage.

TASK 2 – VECTOR REPRESENTATION AND MATRIX OPERATIONS

a) Vector formulation

Represent a normalized patient record as $x = [0.50, 0.60, 0.70, 0.80]^T$, where the entries represent age, glucose, blood pressure and cholesterol after scaling. Let the corresponding importance weights be $w = [0.10, 0.30, 0.35, 0.25]^T$. The preliminary weighted risk score is the dot product: $S = w^T x$.

b) Numerical calculation and interpretation

$S = (0.10 \times 0.50) + (0.30 \times 0.60) + (0.35 \times 0.70) + (0.25 \times 0.80)$

$S = 0.05 + 0.18 + 0.245 + 0.20 = 0.675$.

The illustrative score is 0.675 on a 0-to-1 scale. A higher score indicates a higher weighted value under the selected weights. The weights are assumed for demonstration; a real clinical risk score must be trained, validated and approved using appropriate medical evidence.

TASK 3 – PROBABILITY AND CONDITIONAL PROBABILITY

Let D denote the event that a patient has the disease, and G denote the event that the patient has elevated glucose.

Given: total patients = 1,000; diseased = 200; diseased with elevated glucose = 160; non-diseased = 800; non-diseased with elevated glucose = 120.

a) Probability of elevated glucose

Total patients with elevated glucose = 160 + 120 = 280.

$P(G) = 280/1000 = 0.28 = 28\%$.

b) Probability of disease given elevated glucose

$P(D \mid G) = P(D \cap G) / P(G) = 160/280 = 0.5714 \approx 57.14\%$.

c) Probability of elevated glucose given disease

$P(G \mid D) = 160/200 = 0.80 = 80\%$.

d) Healthcare interpretation

Conditional probability helps estimate how common a condition is within a subgroup. Here, 57.14% of patients with elevated glucose have the disease in this sample. This information can help a hospital prioritize further assessment, but it does not establish a diagnosis. Other clinical information and confirmatory tests are required.

TASK 4 – BAYESIAN STATISTICS FOR DISEASE CLASSIFICATION

Given: $P(D) = 0.10$, $P(\text{not } D) = 0.90$, $P(+ \mid D) = 0.90$ and $P(+ \mid \text{not } D) = 0.05$.

a) Bayes' theorem

$P(D \mid +) = [P(+ \mid D) \times P(D)] / P(+)$

$P(+) = P(+ \mid D)P(D) + P(+ \mid \text{not } D)P(\text{not } D)$

$P(+) = (0.90 \times 0.10) + (0.05 \times 0.90) = 0.09 + 0.045 = 0.135$.

Therefore, $P(D \mid +) = 0.09/0.135 = 0.6667 \approx 66.67\%$.

b) Prior and likelihood

The prior probability $P(D) = 10\%$ represents the disease probability before observing the test result. The likelihood $P(+ \mid D) = 90\%$ describes the probability of a positive result among people who have the disease. The posterior $P(D \mid +) = 66.67\%$ is the updated probability after considering the positive result.

c) Usefulness

Bayesian reasoning combines existing knowledge with new evidence and updates the probability when new information becomes available. It is useful in healthcare because disease prevalence and test accuracy affect how a result should be interpreted. A positive result is not automatically proof of disease.

TASK 5 – ENTROPY AND INFORMATION GAIN

Given 100 records: 40 high-risk and 60 low-risk. Thus, $p(\text{high}) = 0.4$ and $p(\text{low}) = 0.6$.

a) Entropy calculation

$H = -\Sigma\, p_i \log_2(p_i)$

$H = -[0.4 \log_2(0.4) + 0.6 \log_2(0.6)]$

$H \approx -[0.4(-1.3219) + 0.6(-0.7370)] = 0.971$ bits.

b) Interpretation

Entropy measures uncertainty or impurity in the class distribution. The value 0.971 bits is close to the maximum of 1 bit for two equally likely classes, so the dataset has substantial class uncertainty.

c) Information gain

Information gain measures the reduction in entropy after splitting records using a feature. A feature such as glucose level is useful if its split creates groups that are more class-homogeneous. The feature with the greatest information gain can be preferred when constructing a decision tree.

TASK 6 – NUMERICAL COMPUTATION AND DISTANCE MEASUREMENT

Given A = (0.70, 0.80, 0.60) and B = (0.50, 0.75, 0.90), representing glucose, blood pressure and cholesterol.

a) Euclidean distance

$d(A,B) = \sqrt{[(0.70-0.50)^2 + (0.80-0.75)^2 + (0.60-0.90)^2]}$

$= \sqrt{[0.20^2 + 0.05^2 + (-0.30)^2]}$

$= \sqrt{[0.0400 + 0.0025 + 0.0900]} = \sqrt{0.1325} \approx 0.3640$.

b) Similarity interpretation

Euclidean distance measures the straight-line separation between two feature vectors. A smaller distance indicates that the patients have more similar values across the selected normalized features. A larger distance indicates greater difference.

c) Need for normalization

Normalization is needed because the original measurements may use different ranges and units. Without it, a feature with a large numerical scale could dominate the distance and distort the similarity comparison.

TASK 7 – BIAS–VARIANCE TRADE-OFF

a) Identify the models

- Model A: High training error and high validation error indicate high bias (underfitting).
- Model B: Very low training error but high validation error indicate high variance (overfitting).
- Model C: Low and similar training and validation errors indicate good generalization.

b) Effect of model complexity

A model with low complexity may be too simple to capture the real relationships, causing high bias. Increasing complexity can reduce bias, but excessive complexity may fit noise in the training data and increase variance. The objective is to balance both so that the model performs well on unseen patients.

c) Improvements

For Model A, increase model capacity appropriately, add informative features, or reduce excessive regularization. For Model B, use regularization, simplify the model, obtain more representative training data, use cross-validation, and apply early stopping where appropriate.

TASK 8 – OVERFITTING, UNDERFITTING AND MODEL CAPACITY

a) Model behaviour

- The simple model that fails to capture disease patterns is underfitting.
- The moderately complex model with consistent performance is generalizing appropriately.
- The highly complex model with near-perfect training performance but poor unseen-patient performance is overfitting.

b) Model capacity

Model capacity is the ability of a model to represent different patterns and relationships in data. Low-capacity models may not learn complex relationships. Higher-capacity models can represent more complex patterns.

c) Why performance can later decline

As capacity increases from too low, the model can capture meaningful patterns and performance may improve. Beyond an appropriate level, it may also learn noise and accidental details in the training set. Training error may continue to decrease while validation error increases, which is overfitting. Validation data and regularization help select an appropriate capacity.

TASK 9 – SUPERVISED LEARNING AND MODEL EVALUATION

a) Why supervised learning

The hospital has historical patient records with known labels (diseased or disease-free). A supervised learning algorithm learns the relationship between input features and these target labels, then predicts the label for a new patient.

b) Training and testing sets

The training set is used to fit the model and learn its parameters. The testing set is kept separate and is used after training to estimate how well the model performs on unseen records. A validation set may also be used for model selection and hyperparameter tuning.

c) Accuracy

Accuracy = (Number of correct predictions / Total test predictions) × 100

= (92/100) × 100 = 92%.

The model correctly classified 92 out of 100 test patients. Accuracy alone may not be sufficient for healthcare, especially when classes are imbalanced; sensitivity, specificity, precision, recall and F1-score should also be considered.

TASK 10 – UNSUPERVISED LEARNING AND PATIENT GROUPING

a) Suitability

The records do not contain predefined risk labels. Unsupervised learning is suitable because it can discover structure or groups in the patient data without requiring target labels.

b) K-means procedure

- Choose the number of clusters K.
- Initialize K cluster centroids.
- Calculate the distance from each patient record to each centroid and assign the record to its nearest centroid.
- Recalculate each centroid as the mean of the records assigned to that cluster.
- Repeat assignment and centroid updates until assignments or centroids stop changing significantly.

c) Distance-based similarity

K-means commonly uses Euclidean distance. Patients with smaller distances in the selected normalized feature space are assigned to the same cluster, indicating similar measured characteristics.

d) Clinical use

Clusters may reveal groups with similar age, blood pressure, glucose and cholesterol profiles. Clinicians can investigate these groups further, identify patterns for research or plan additional screening. Clusters are exploratory and must not replace medical diagnosis.

TASK 11 – COMPARISON OF SUPERVISED AND UNSUPERVISED LEARNING

a) Suitable learning paradigm

Dataset A contains disease labels, so supervised learning is appropriate for learning to predict disease status. Dataset B has no labels, so unsupervised learning is appropriate for discovering patient groups or hidden patterns.

b) Objectives and outputs

Aspect	Supervised learning	Unsupervised learning
Input	Features with known labels	Features without labels
Objective	Learn mapping from features to target	Discover structure or groups
Output	Predicted class or numerical value	Clusters, associations or representations
Example	Disease / disease-free classification	K-means patient grouping

c) Recommendation

For predicting disease status in a new patient, use a supervised classification model trained on labelled and representative patient records. Unsupervised clustering can complement it by exploring unlabelled records and identifying patient subgroups. Predictions should be validated and interpreted by qualified healthcare professionals.

TASK 12 – DESIGN CHALLENGE: SELECTING AND ADAPTING A LEARNING STRATEGY

A. Impact Analysis

a) Impact of increasing dataset size

Increasing the dataset from 10,000 to 1,000,000 records increases storage, memory and processing requirements. The data representation must support efficient loading and processing, and numerical computations such as normalization, distance calculation and model training must scale to the larger number of records. Training may take longer, so efficient algorithms, batching, parallel processing and suitable hardware may be needed.

b) Data quality and variability

Normalization prevents differences in feature scales from distorting learning. Missing values should be examined and handled using an appropriate method, such as median imputation for numerical features, a separate missing category for selected categorical features, or removal when justified. Data variability is important because the training data should represent the range of patients the system will encounter. Poor-quality or unrepresentative data can reduce reliability.

B. Evaluation of the Original Approach

a) Is one supervised approach sufficient?

A single supervised classifier can predict disease risk when labels are available, but it does not directly discover groups in unlabelled records. Therefore, it is not sufficient for both prediction and patient-group discovery.

b) Limitations of labelled learning

A supervised model learns from the labels supplied to it. It may not use the full structure of unlabelled records, even though those records may contain useful patterns. It can also inherit label errors, class imbalance and sampling bias from the labelled dataset.

C. Adaptation or Redesign

a) Combined strategy

Use a two-part analytical pipeline. First, clean the dataset, identify missing values, encode categorical variables where applicable, and normalize numerical features. Next, train a supervised classifier on labelled records to predict disease status. In parallel, apply an unsupervised clustering algorithm such as K-means to normalized patient features, including suitable unlabelled records, to discover patient groups. Compare cluster profiles and investigate them clinically. The two outputs should be treated as complementary: classification predicts the target, while clustering explores structure.

b) Incorporating mathematical concepts

- Probability: estimate disease probabilities and interpret classification confidence.
- Distance measures: use Euclidean or another suitable distance for patient similarity and clustering.
- Normalization: scale numerical features before distance-based learning and gradient-based optimization.
- Model capacity: select a classifier with enough capacity to learn useful patterns without overfitting; use validation and regularization.
- Missing values: impute or otherwise handle missing observations before model fitting, and apply the same preprocessing consistently.

D. Evidence-Based Defence

a) Experiments and evaluation

- Compare the original supervised-only approach with the combined classification-and-clustering strategy using the same data splits and preprocessing rules.
- For classification, evaluate accuracy, precision, recall/sensitivity, specificity, F1-score, ROC-AUC and a confusion matrix. Pay particular attention to false negatives because missed high-risk patients can be consequential.
- For clustering, evaluate silhouette score or another suitable internal clustering measure, inspect cluster sizes and feature profiles, and assess whether groups are stable and clinically interpretable.
- Measure training time, prediction time, memory use and scalability as the dataset grows.
- Use a held-out test set for final classification evaluation and cross-validation on the training data for model selection. Avoid allowing test data to influence preprocessing or tuning.

b) Final justification

The redesigned hybrid strategy is more suitable because it addresses both required goals: predicting disease risk from labelled records and discovering groups in unlabelled records. Normalization supports fair distance calculations, missing-value handling improves data usability, probability supports risk interpretation, and capacity control helps generalization. Efficient processing methods can address the larger dataset. The final system should be treated as a decision-support tool, with clinical validation, privacy safeguards and professional oversight before deployment.

CONCLUSION

The mathematical foundations of linear algebra, probability, information theory and numerical computation directly influence the reliability of machine learning systems. Normalization supports meaningful comparisons and stable optimization; probability and Bayes' theorem help interpret uncertain evidence; entropy supports feature selection; and distance measures support similarity analysis. The bias–variance trade-off and model capacity explain why models must be selected for generalization rather than training performance alone.

For the hospital scenario, supervised classification is appropriate for disease-risk prediction using labelled records, while unsupervised clustering can reveal patterns in unlabelled records. A combined, carefully evaluated approach is therefore recommended for the expanded one-million-record dataset. The system should support, not replace, clinical judgement.

REFERENCE

Assignment source: CSA47 – Deep Learning, Assignment 1, CO1 – Mathematical Foundations and Machine Learning Principles, Saveetha School of Engineering. The questions, scenario and task structure in this report are based on the supplied assignment sheet.